

HW

$$\text{Info}(D) = I(9,5) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.940$$

$$\text{Info}_{\text{age}}(D) = \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2)$$

$$= \frac{5}{14} \left(-\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) \right) + \frac{4}{14} \left(-\frac{4}{4} \log_2\left(\frac{4}{4}\right) - \frac{0}{4} \log_2\left(\frac{0}{4}\right) \right) + \frac{5}{14} \left(-\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \right)$$

$$= 0.346768 + 0 + 0.346768 = 0.6935$$

$$\text{Info}_{\text{income}}(D) = \frac{4}{14} I(3,1) + \frac{6}{14} I(4,2) + \frac{4}{14} I(2,2)$$

low (3,1) medium (4,2) high (2,2)

$$= \frac{4}{14} \left(-\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) \right) + \frac{6}{14} \left(-\frac{4}{6} \log_2\left(\frac{4}{6}\right) - \frac{2}{6} \log_2\left(\frac{2}{6}\right) \right) + \frac{4}{14} \left(-\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right) \right)$$

$$= 0.231794 + 0.393555 + 0.285714 = 0.911063$$

$$\text{Info}_{\text{student}}(D) = \frac{7}{14} I(6,1) + \frac{7}{14} I(3,4)$$

$$= \frac{7}{14} \left(-\frac{6}{7} \log_2\left(\frac{6}{7}\right) - \frac{1}{7} \log_2\left(\frac{1}{7}\right) \right) + \frac{7}{14} \left(-\frac{3}{7} \log_2\left(\frac{3}{7}\right) - \frac{4}{7} \log_2\left(\frac{4}{7}\right) \right)$$

$$= 0.295836 + 0.492614 = 0.78845$$

Yes (6,1) No (3,4)

$$\text{Info}_{\text{credit}}(D) = \frac{8}{14} I(6,2) + \frac{6}{14} I(3,3)$$

$$= \frac{8}{14} \left(-\frac{6}{8} \log_2\left(\frac{6}{8}\right) - \frac{2}{8} \log_2\left(\frac{2}{8}\right) \right) + \frac{6}{14} \left(-\frac{3}{6} \log_2\left(\frac{3}{6}\right) - \frac{3}{6} \log_2\left(\frac{3}{6}\right) \right)$$

$$= 0.463587 + 0.428571 = 0.892158$$

fair (6,2) excellent (3,3)

$$\text{Gain}(\text{age}) = 0.940 - 0.6935 = 0.2465$$

$$\text{Gain}(\text{income}) = 0.940 - 0.911063 = 0.028937$$

$$\text{Gain}(\text{student}) = 0.940 - 0.78845 = 0.15155$$

$$\text{Gain}(\text{Credit}) = 0.940 - 0.892158 = 0.047842$$

∴ Gain (age) มากที่สุด

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Training data set: Who buys computer?

age	income	student	credit rating	buys computer
≤30	high	no	fair	no
≤30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤30	medium	no	fair	no
≤30	low	yes	fair	yes
>40	medium	yes	fair	yes
≤30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
>40	high	yes	fair	yes
>40	medium	no	excellent	no

Note: The data set is adapted from "Playing Tennis" example of R. Quinlan